

Credit Card Approval Prediction System

Project Description

The Credit Card Approval Prediction System is a Machine Learning-based web application that predicts whether a customer's credit card application will be approved or rejected based on their financial and personal information.

The system uses classification algorithms such as:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier

The application is developed using Python and Flask and provides a simple user interface where users can enter applicant details and receive instant predictions.

Scenarios

Scenario 1

A working professional with a high annual income, graduate education, and stable employment applies for a credit card. The system predicts that the application is approved.

Scenario 2

A pensioner with low annual income and multiple loans applies for a credit card. The system predicts that the application is rejected.

Scenario 3

A commercial associate with moderate income and good financial status applies for a credit card. The system predicts the approval status in real time.

Technical Architecture

The project follows a three-layer architecture:

Frontend

- HTML
- CSS
- Bootstrap

Backend

- Flask Framework
- Python

Machine Learning Layer

- Pandas
- NumPy
- Scikit-Learn
- XGBoost

Deployment

- Render Cloud Platform
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Pre-Requisites

- Python Programming
 - Machine Learning Fundamentals
 - Flask Framework
 - HTML and CSS
 - Data Visualization
 - Git and GitHub
 - Cloud Deployment using Render
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Project Workflow

Milestone 1: Dataset Collection and Analysis

- Download Credit Card Approval Dataset.
- Understand dataset features.
- Analyze approval patterns.

Milestone 2: Data Preprocessing

- Handle missing values.
- Remove duplicates.
- Encode categorical features.
- Prepare training dataset.

Milestone 3: Model Development

- Train Logistic Regression.
- Train Decision Tree.
- Train Random Forest.
- Train XGBoost.
- Compare model accuracy.

Milestone 4: Flask Application Development

- Design web interface.
- Create prediction form.
- Display prediction results.

Milestone 5: Deployment

- Deploy application on Render.
- Test online accessibility.

Milestone 6: Conclusion

- Evaluate system performance.
 - Discuss future enhancements.
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Exploring the Website Pages

Home Page

Displays project information and prediction form.

Prediction Page

Allows the user to enter:

- Gender
- Annual Income
- Income Type
- Education
- Family Status
- Number of Loans
- Family Members

Result Page

Displays:

- Credit Card Approved
or
- Credit Card Rejected

Conclusion

The Credit Card Approval Prediction System successfully predicts credit card approval decisions using Machine Learning techniques. The project demonstrates practical implementation of data preprocessing, feature engineering, classification algorithms, and web deployment.

The Random Forest and XGBoost models achieved an accuracy close to 99–100% on the generated dataset. The Flask web application provides an easy-to-use interface for real-time predictions.

Future improvements may include:

- Using real banking datasets
- Adding user authentication
- Storing prediction history in a database
- Deploying with Docker and CI/CD pipelines
- Integrating explainable AI techniques for prediction transparency